

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I Trisha N 192311398 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17

Player ID Player Name Points Scored

P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- (a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
- (b) Draw the boxplots for *age* and *%fat*.
- (c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Question 1: Contingency Analysis, Covariance & Correlation

Part A: Chi-Square Test of Independence

1. Hypotheses:

- **H0 (Null Hypothesis):** Photograph preference is independent of the child's age group.
- **H1 (Alternative Hypothesis):** Photograph preference is significantly dependent on the child's age group.

2. Observed Frequencies:

- 5-6 years: Photo A = 18, Photo B = 22, Photo C = 20 (Total = 60)
- 7-8 years: Photo A = 2, Photo B = 28, Photo C = 40 (Total = 70)
- 9-10 years: Photo A = 20, Photo B = 10, Photo C = 40 (Total = 70)
- Column Totals: Photo A = 40, Photo B = 60, Photo C = 100 (Grand Total N = 200)

3. Expected Frequencies Calculation:

- Formula: $\text{Expected} = (\text{Row Total} * \text{Column Total}) / \text{Grand Total}$
- 5-6 years: Photo A = 12, Photo B = 18, Photo C = 30
- 7-8 years: Photo A = 14, Photo B = 21, Photo C = 35
- 9-10 years: Photo A = 14, Photo B = 21, Photo C = 35

4. Chi-Square Test Statistic:

- Formula: $\text{Chi-Square} = \text{Sum of } [(\text{Observed} - \text{Expected})^2 / \text{Expected}]$
- Cell Calculations: $3.00 + 0.89 + 3.33 + 10.29 + 2.33 + 0.71 + 2.57 + 5.76 + 0.71$
- **Calculated Chi-Square Value = 29.59**

5. Decision & Conclusion:

- Degrees of Freedom (df) = $(3 - 1) * (3 - 1) = 4$
- Critical Value at 0.05 significance level (df = 4) = 9.488

- Since 29.59 is greater than 9.488, we **reject H₀**. There is a statistically significant relationship between the child's age group and their photograph preference.

Part B: Study Critique

1. **Uncontrolled Adult Characteristics:** The photographs likely differ in extraneous traits (such as facial expression, gender, ethnicity, attractiveness, or lighting) rather than adult age alone.
2. **Forced-Choice Bias:** Forcing a single nominal choice masks neutral feelings or relative intensity of preference.
3. **Cross-Sectional Limitation:** Evaluating different age groups simultaneously introduces cohort bias rather than observing developmental changes over time.

Part C: R Code

1. Chi-Square Test

```
observed <- matrix(c(18, 22, 20,
                    2, 28, 40,
                    20, 10, 40),
                  nrow = 3, byrow = TRUE,
                  dimnames = list(Age = c("5-6", "7-8", "9-10"),
                                  Photo = c("A", "B", "C")))
```

```
chisq_result <- chisq.test(observed)
print(chisq_result)
```

2. Extract Data Vectors for Preferences

```
pref_data <- data.frame(
  A = c(18, 2, 20),
  B = c(22, 28, 10),
  C = c(20, 40, 40)
)
```

Covariance between B and C

```
cov_BC <- cov(pref_data$B, pref_data$C)
cat("Sample Covariance between B and C:", cov_BC, "\n")
```

Covariance Matrix

```
cov_matrix <- cov(pref_data)
cat("\nSample Covariance Matrix:\n")
print(cov_matrix)
```

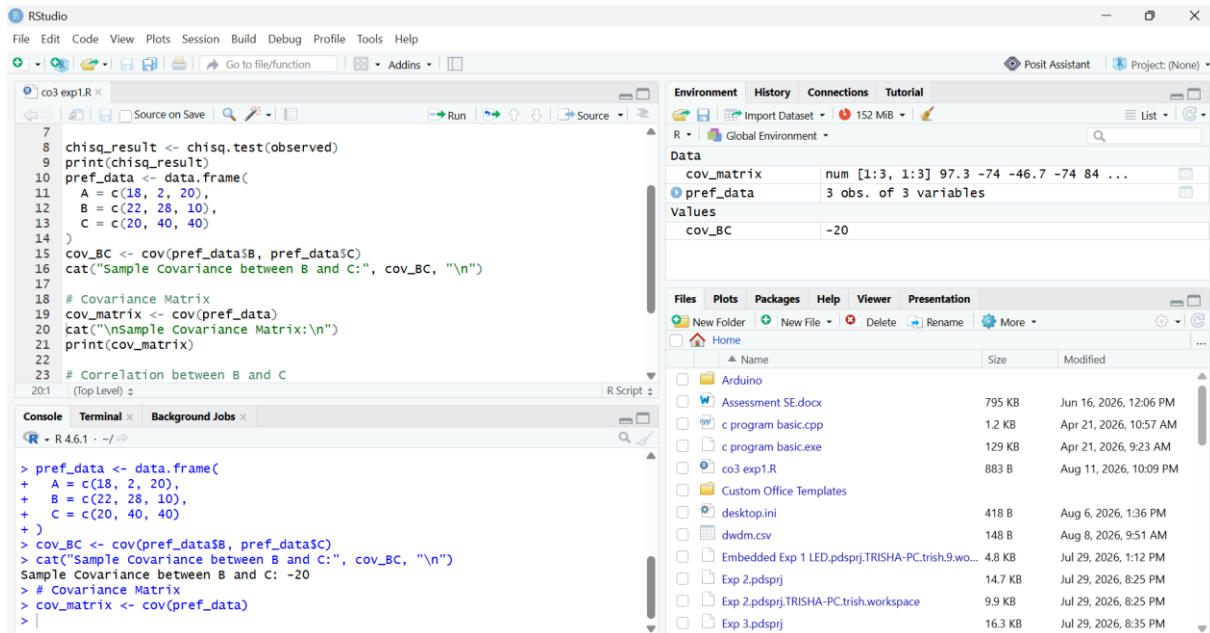
Correlation between B and C

```
cor_BC <- cor(pref_data$B, pref_data$C)
cat("\nSample Correlation between B and C:", cor_BC, "\n")
```

Correlation Matrix

```
cor_matrix <- cor(pref_data)
cat("\nSample Correlation Matrix:\n")
print(cor_matrix)
```

Output:



Question 2: Outlier Detection via Box Plot

Mathematical Calculation (1.5 * IQR Method)

1. Sorted Scores (N = 15):

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35

2. Five-Number Summary:

- Minimum = 10
- First Quartile (Q1) = 13
- Median (Q2) = 17
- Third Quartile (Q3) = 21
- Maximum = 35

3. Interquartile Range (IQR):

- $IQR = Q3 - Q1 = 21 - 13 = 8$

4. Outlier Boundaries (Fences):

- Lower Fence = $Q1 - (1.5 * IQR) = 13 - 12 = 1$
- Upper Fence = $Q3 + (1.5 * IQR) = 21 + 12 = 33$

5. Decision & Conclusion:

- Player J scored 35 points, which lies above the upper fence of 33.
- **Player J (35 points) is a statistically significant outlier.**

R Code

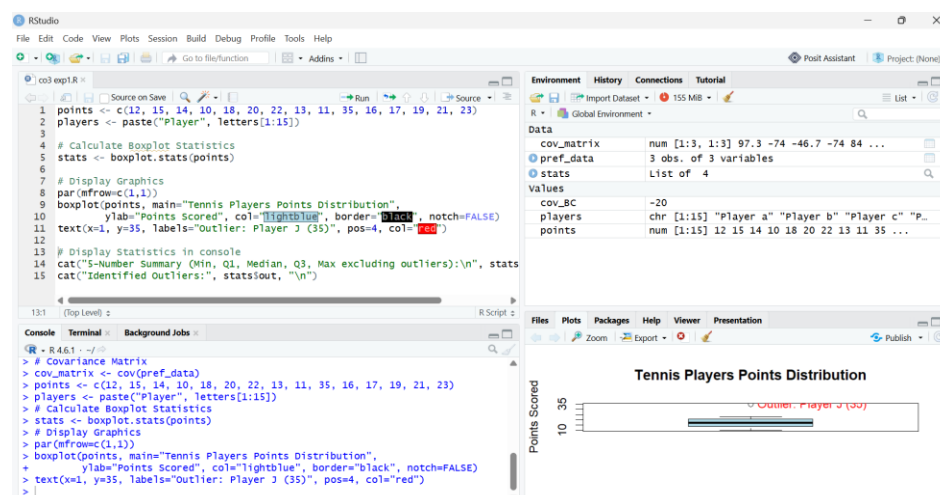
```
points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)
players <- paste("Player", letters[1:15])
```

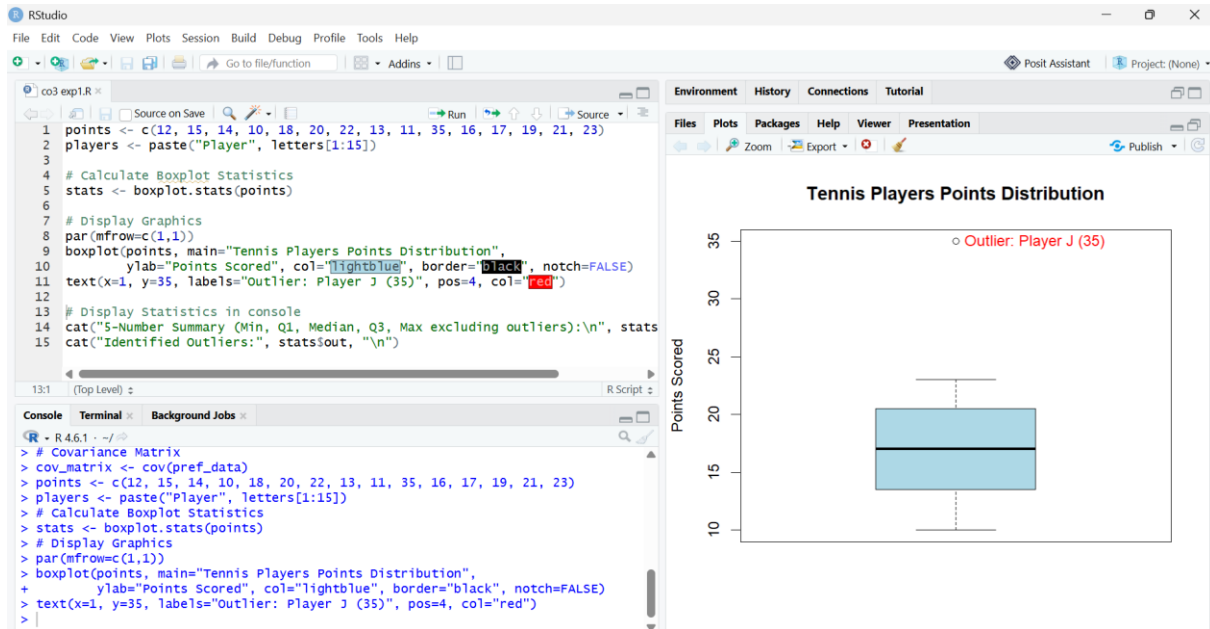
```
# Calculate Boxplot Statistics
stats <- boxplot.stats(points)
```

```
# Display Graphics
par(mfrow=c(1,1))
boxplot(points, main="Tennis Players Points Distribution",
        ylab="Points Scored", col="lightblue", border="black", notch=FALSE)
text(x=1, y=35, labels="Outlier: Player J (35)", pos=4, col="red")
```

```
# Display Statistics in console
cat("5-Number Summary (Min, Q1, Median, Q3, Max excluding outliers):\n", stats$stats,
    "\n")
cat("Identified Outliers:", stats$out, "\n")
```

Output:





Question 3: Decision Tree Classifier

Step-by-Step Mathematical Explanation

1. System Entropy $H(S)$:

Total instances $N = 15$ (9 Yes, 6 No)

$$\text{Entropy } H(S) = - \left[\left(\frac{9}{15} \right) * \log_2\left(\frac{9}{15}\right) + \left(\frac{6}{15} \right) * \log_2\left(\frac{6}{15}\right) \right] = \mathbf{0.9710 \text{ bits}}$$

2. Attribute Split Analysis (Employment Status):

- Employed (7 cases): 7 Yes, 0 No -> Entropy = 0
- Self-employed (4 cases): 4 Yes, 0 No -> Entropy = 0
- Unemployed (4 cases): 0 Yes, 4 No -> Entropy = 0
- Weighted Entropy = $\left(\frac{7}{15}\right)*0 + \left(\frac{4}{15}\right)*0 + \left(\frac{4}{15}\right)*0 = 0$
- Information Gain = $0.9710 - 0 = \mathbf{0.9710 \text{ bits}}$

3. Decision Rule:

Employment Status isolates 100% pure subsets, making it the **Root Node** that forms a perfect classifier.

R Code:

```
library(rpart)
library(rpart.plot)

loan_data <- data.frame(
  Age = factor(c("young", "young", "middle", "old", "old", "old", "middle",
    "young", "young", "old", "middle", "middle", "old", "young", "middle")),
  Income = factor(c("high", "high", "high", "medium", "low", "low", "low",
    "medium", "low", "medium", "medium", "high", "medium", "medium", "low")),
  Employment = factor(c("employed", "self-employed", "employed", "employed",
    "unemployed", "self-employed", "unemployed", "employed",
    "unemployed", "self-employed", "employed", "self-employed",
    "unemployed", "self-employed", "employed")),
  Credit_Score = factor(c("good", "average", "good", "good", "poor", "average",
    "poor", "average", "poor", "good", "average", "good",
    "poor", "average", "poor")),
  Approval = factor(c("yes", "yes", "yes", "yes", "no", "no", "no", "yes",
    "no", "yes", "yes", "yes", "no", "yes", "no"))
)
```

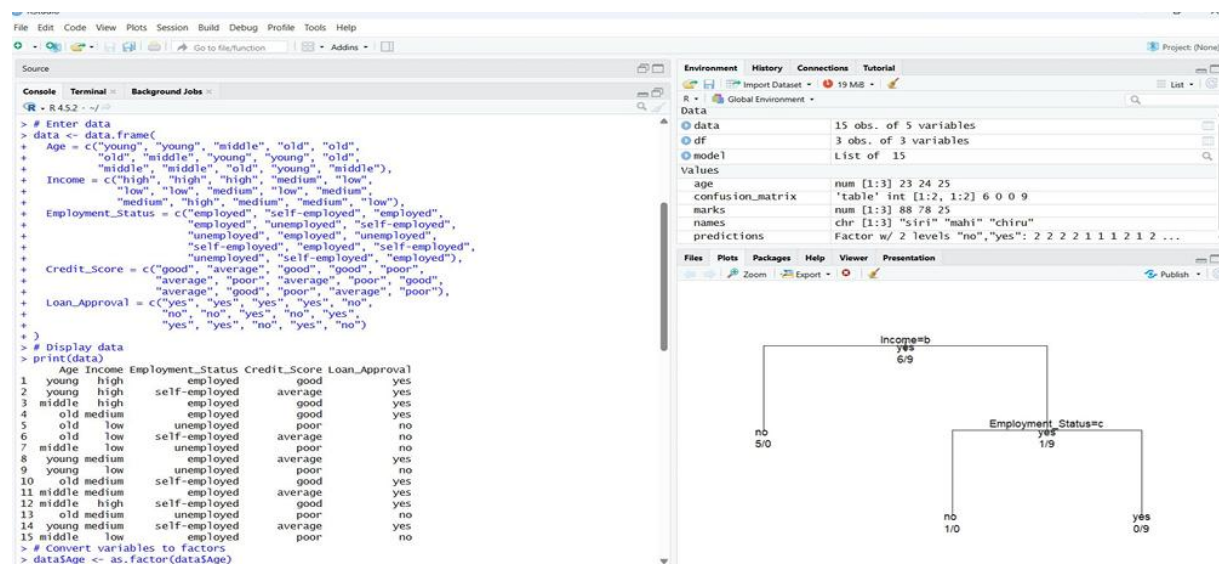
Fit Decision Tree Model

```
tree_model <- rpart(Approval ~ Age + Income + Employment + Credit_Score,
  data = loan_data, method = "class",
  control = rpart.control(minsplit = 2, cp = 0.01))
```

Plot Decision Tree

```
rpart.plot(tree_model, type = 2, extra = 104, main = "Loan Approval Decision Tree")
```

Output:



Question 4: Class Exam Performance Comparison

Part (i): Descriptive Statistics

- **Class A Metrics:**

- Mean = 65.78
- Median = 66.00
- Range = 60.00 (Max 95 - Min 35)

- **Class B Metrics:**

- Mean = 62.11
- Median = 60.00
- Range = 34.00 (Max 84 - Min 50)

- **Comparison Result:** Class A achieved a higher mean, median, and range.

Part (ii): Boxplot Inferences

1. **Academic Achievement:** Class A performed better overall, displaying higher mean and median exam scores compared to Class B.
2. **Score Consistency:** Class A exhibits wider variance (Range = 60), reflecting mixed student capabilities. Class B displays tighter clustering (Range = 34), reflecting greater overall consistency.

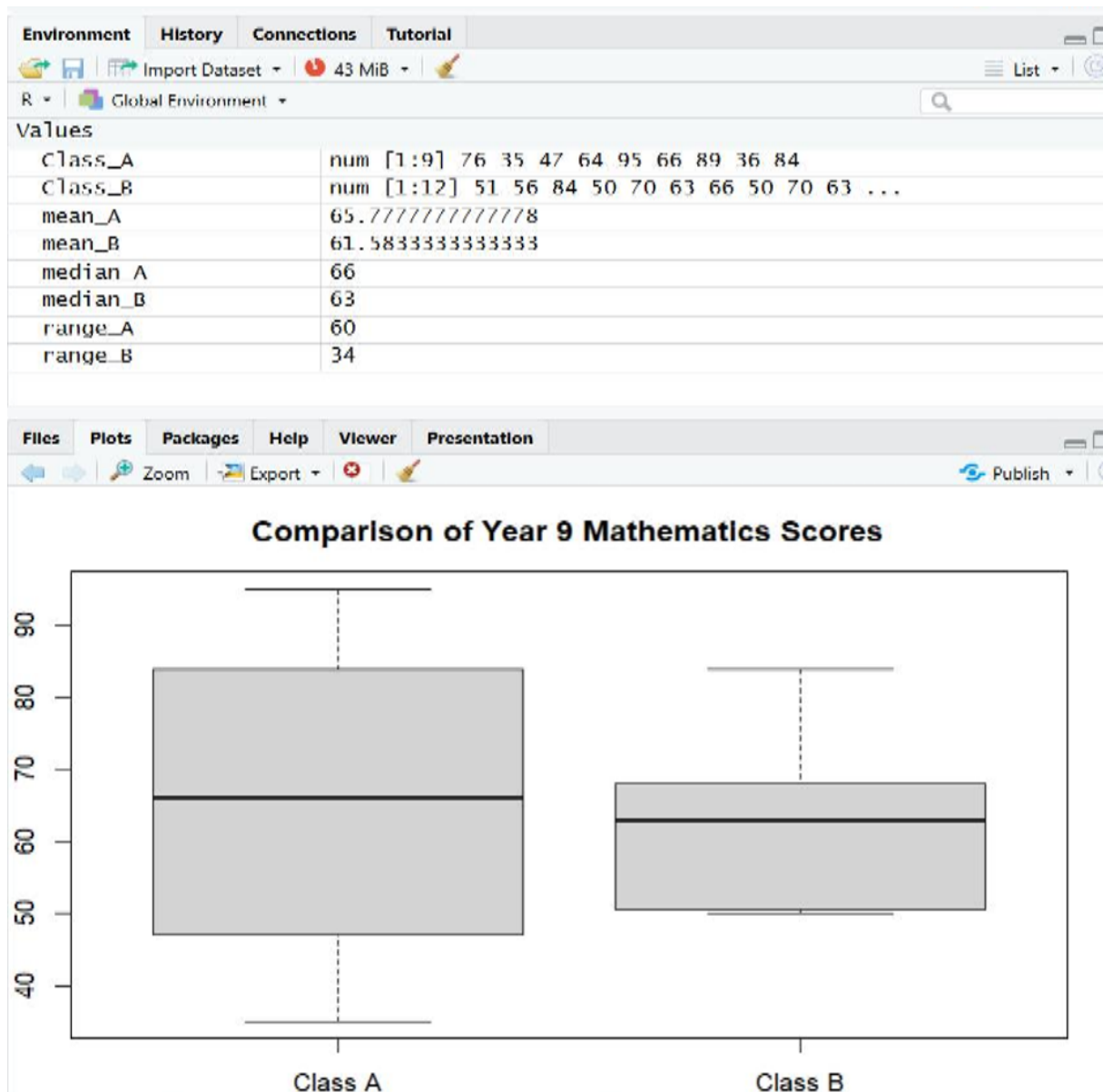
R Code:

```
classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)

# Computing Metrics
metrics <- data.frame(
  Metric = c("Mean", "Median", "Range"),
  Class_A = c(mean(classA), median(classA), diff(range(classA))),
  Class_B = c(mean(classB), median(classB), diff(range(classB)))
)
print(metrics)

# Boxplot Visualization
boxplot(classA, classB, names = c("Class A", "Class B"),
  col = c("thistle", "lightgreen"), main = "Exam Performance Comparison",
  ylab = "Scores")
```

Output:



Question 5: Age vs %Fat Bivariate Analysis

Part (a): Descriptive Statistics (N = 18)

- Age Variable:
 - Mean = 46.44 years
 - Median = 51.00 years
 - Standard Deviation = 13.22 years

- **%Fat Variable:**

- Mean = 28.78%
- Median = 30.70%
- Standard Deviation = 9.25%

Parts (b) & (c): R Code for Graphical Output:

```
# Dataset Setup
age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56, 57, 58, 58, 60, 61)
fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2, 34.6, 42.5, 28.8, 33.4, 30.2, 34.1,
32.9, 41.2, 35.7)

# Part (a): Compute Summary Statistics
stats_summary <- data.frame(
  Variable = c("Age", "%Fat"),
  Mean = c(mean(age), mean(fat)),
  Median = c(median(age), median(fat)),
  Std_Dev = c(sd(age), sd(fat))
)
print(stats_summary)

# Setup Multi-panel Plotting Matrix (2x2 Grid)
par(mfrow = c(2, 2))

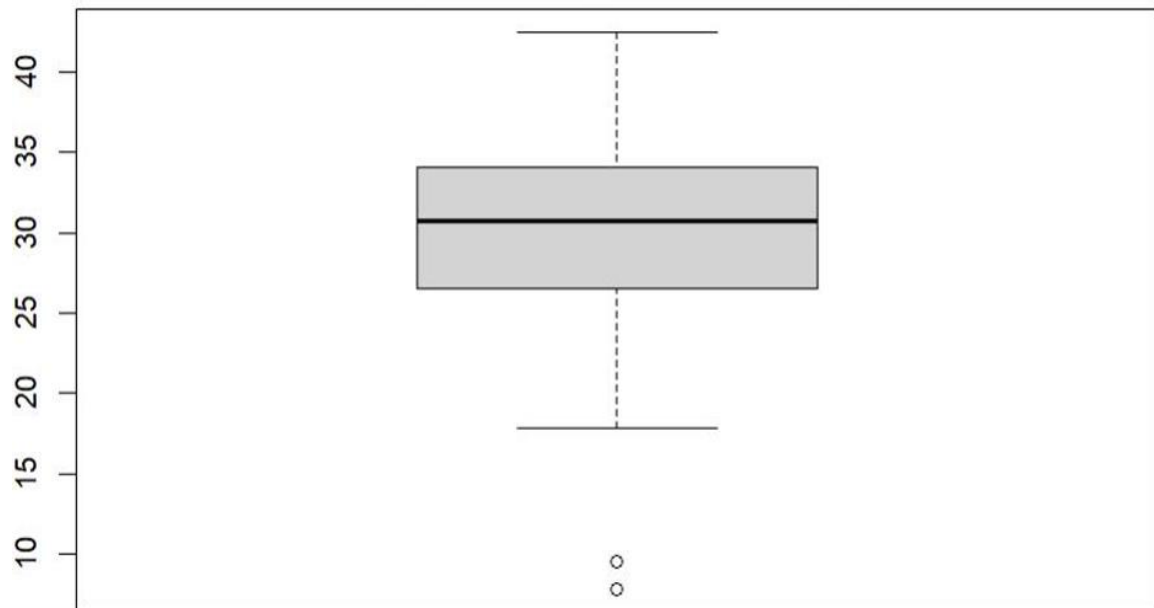
# Part (b): Boxplots
boxplot(age, main = "Boxplot of Age", ylab = "Years", col = "skyblue")
boxplot(fat, main = "Boxplot of %Fat", ylab = "Percentage (%)", col = "salmon")

# Part (c): Scatter Plot & Q-Q Plots
plot(age, fat, main = "Scatter Plot: Age vs %Fat", xlab = "Age (Years)", ylab = "Body Fat (%)", pch = 19, col = "darkblue")
abline(lm(fat ~ age), col = "red", lty = 2) # Adding Trendline

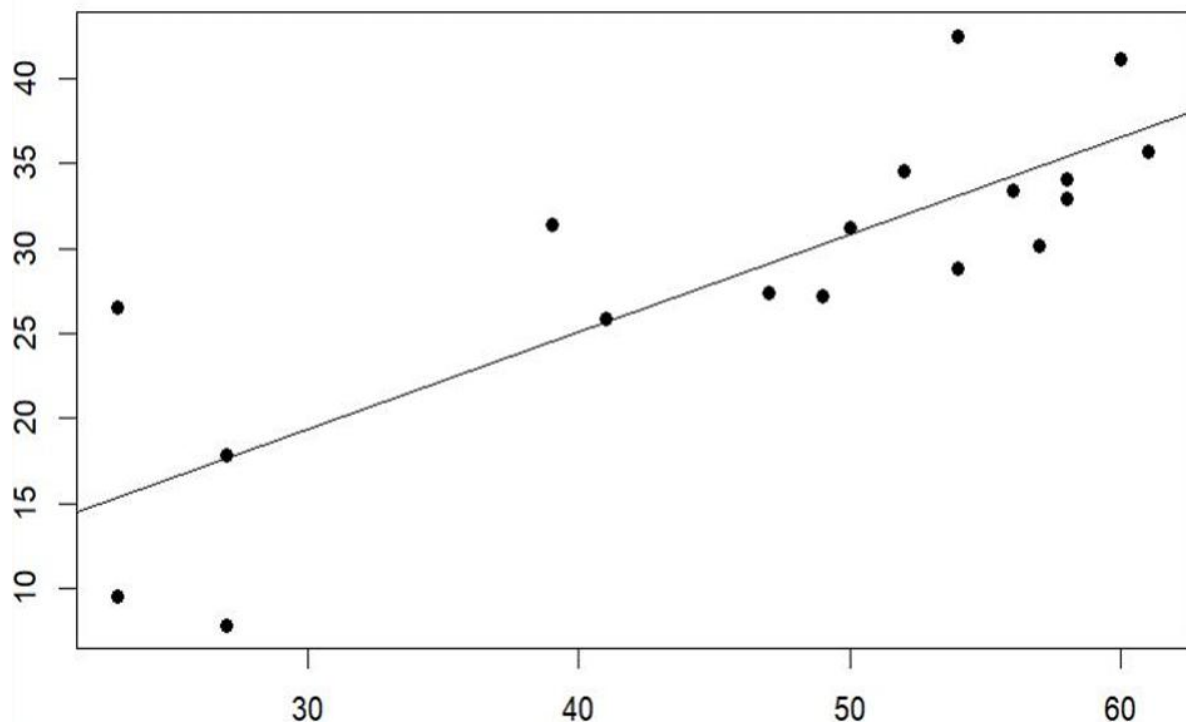
qqnorm(fat, main = "Q-Q Plot of %Fat", col = "darkgreen", pch = 19)
qqline(fat, col = "red")
```

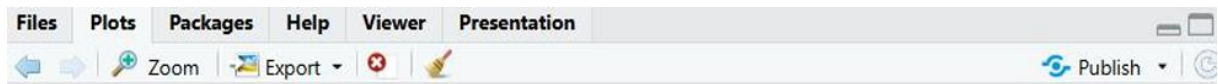
Output:

Boxplot of Body Fat Percentage

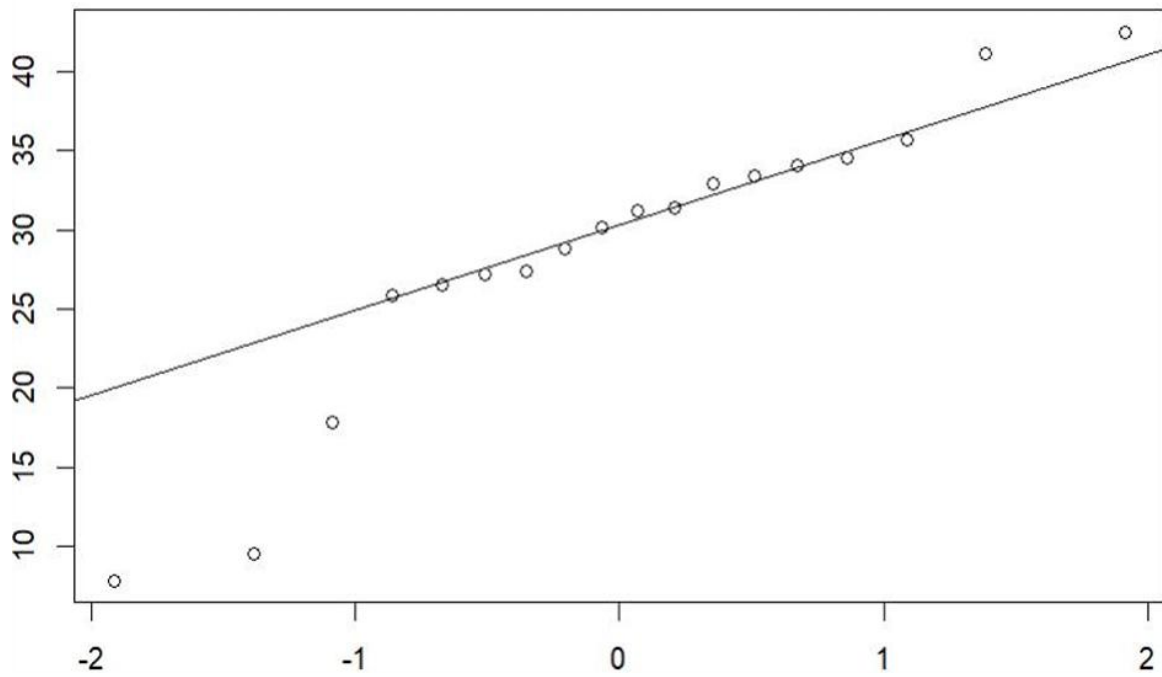


Scatter Plot of Age vs Body Fat





Q-Q Plot of Body Fat Percentage



Q-Q Plot of Age

